

# **Bremelanotide (PT 141)**

## **1. Drug Name**

- **Generic:** Bremelanotide
- **Brand Name:** Vyleesi®

## **2. Drug Class**

- **Melanocortin Receptor Agonist (MC4R agonist)**

## **3. Mechanism of Action**

**Bremelanotide is an agonist of melanocortin receptors, primarily MC4R, involved in regulating sexual desire and arousal. Its exact mechanism in treating hypoactive sexual desire disorder (HSDD) is not fully understood, but it's thought to enhance sexual desire via central nervous system pathways without directly affecting hormones like estrogen or testosterone.**

## **4. Indications**

- **FDA-approved: Treatment of acquired, generalized hypoactive sexual desire disorder (HSDD) in premenopausal women**
  - **Acquired:** Develops in a patient who previously had no issues with sexual desire.
  - **Generalized:** Not limited to certain types of partners, situations, or stimuli.

## **5. Dosage and Administration**

- **Dose: 1.75 mg subcutaneously (SC) in the abdomen or thigh**
- **Timing: As needed, approximately 45 minutes before anticipated sexual activity**
- **Maximum Frequency:**
  - **No more than 1 dose per 24 hours**
  - **No more than 8 doses per month**
- **Nasal Spray: 5mg/ml, 5ml**
  - **1-2 pumps intranasally one hour prior to sexual activity**

### **Dosage Adjustment:**

- **No adjustment needed for renal or hepatic impairment.**
- **Discontinue after 8 weeks if no improvement in sexual desire.**

## **6. Contraindications**

- **Uncontrolled hypertension**
- **Known cardiovascular disease (due to potential for transient BP elevation)**

## **7. Warnings and Precautions**

- **Blood Pressure Increases:** May cause transient increases in blood pressure and decrease heart rate; monitor if pre-existing hypertension.
- **Nausea:** Common; can lead to discontinuation.
- **Darkening of Skin (Hyperpigmentation):** More common in patients with darker skin tones or with frequent use.

## **8. Adverse Reactions**

### **Common ( $\geq 2\%$ ):**

- **Nausea (40%)**
- **Flushing**
- **Injection site reactions**
- **Headache**
- **Vomiting**
- **Fatigue**

### **Serious:**

- **Hypertension**
- **Bradycardia**
- **Fainting (rare)**

## **9. Drug Interactions**

- **Naltrexone:** Reduces naltrexone exposure (CYP-mediated interaction); avoid concurrent use.
- **Caution with medications that affect blood pressure or heart rate.**

## **10. Pharmacokinetics**

- **Absorption:** Rapid after SC injection; peak plasma concentration ~1 hour
- **Bioavailability:** ~100% via SC route
- **Metabolism:** Primarily by hydrolysis and oxidation (non-CYP pathways)
- **Half-life:** ~2.7 hours
- **Elimination:** Renal and fecal routes

## **11. Use in Special Populations**

- **Pregnancy:** Category not assigned; use only if potential benefit justifies risk
- **Lactation:** Not recommended; unknown if excreted in human milk
- **Pediatrics:** Not indicated
- **Geriatrics:** Not studied

## 12. Storage

- **Store in refrigerator (2°C to 8°C / 36°F to 46°F)**
- **Can be kept at room temperature (20°C to 25°C / 68°F to 77°F) for up to 30 days**
- **Protect from light; do not freeze**

## 13. Supplied As

- **Single-use prefilled autoinjector, Nasal Spray**
- **Strength: 1.75 mg/0.3 mL solution for subcutaneous injection, Nasal Spray 5mg/ml**

"This compounded formulation has not been evaluated or approved by the U.S. Food and Drug Administration. Use is based on clinical judgment and is considered off-label."

"This information is not a substitute for professional medical advice, diagnosis, or treatment. Always seek the advice of your physician or other qualified health provider with any questions you may have regarding a medical condition."